

Tutorial 1

Conceptual and logical database design

Week 3

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Question 1. Modelling a veterinary clinic chain

A chain of veterinary clinics is to be modelled. The company runs several clinics; a clinic is identified by a *clinicID* and has an address (city, street, postcode) and a phone number. Each clinic employs one or more veterinarians; a veterinarian is identified by an *empID* and has a name, a specialisation and a salary, and works at exactly one clinic. Each clinic is managed by exactly one veterinarian, who must be one of the veterinarians working at that clinic, and a veterinarian manages at most one clinic.

Owners register at exactly one clinic. An owner is identified by an *ownerID* and has a name, a phone number and an email address. An owner owns one or more pets; pets are numbered 1, 2, 3, ... within their owner (two owners may both own pet number 1), and each pet has a name, a species and a date of birth. A pet may be treated several times; treatments of a pet are numbered 1, 2, 3, ... chronologically and record a date, a diagnosis and a fee, and each is carried out by exactly one veterinarian. A treatment may use zero, one or several medications; a medication is identified by a *medCode* and has a name and a unit price, and the quantity used in a treatment is recorded.

- (a) List the entity types with their attributes and keys. For each **weak** entity type give its partial key and its identifying relationship, and name the identifying (owner) entity type.
- (b) List the relationship types with their cardinality ratios ($1 : 1 / 1 : N / M : N$) and participation constraints (total / partial), and draw the ER diagram. State every assumption you make.
- (c) Give the relational schema produced by the standard ER-to-relational mapping. Underline every primary key (i.e. write it clinicID), mark every foreign key with an asterisk (i.e. write it clinicID*), and say which relationship types become a relation of their own and why.
- (d) Give two constraints from this application that a relational schema cannot enforce using key and foreign-key constraints alone, and say what mechanism would be needed instead.